Kakeya Skeleton with Residual Turbo Compression: A Rate–Distortion Framework for LLM KV Cache Compression

KakeyaTurbo Contributors
https://github.com/FluffyAIcode/LLM-KV--Cache-compress

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Abstract

The key–value (KV) cache in transformer-based large language models grows linearly with the context length and dominates system throughput at long-context and multi-turn workloads. Existing post-hoc compressors fall into two orthogonal families: *per-vector scalar quantizers* (TurboQuant, SmoothQuant-KV) that exploit coordinate-wise Gaussianity, and *cross-token low-rank decomposers* (H_2O, Scissorhands) that exploit trajectory redundancy. We propose KakeyaTurbo, a unified rate–distortion (RD) codec that constructs a *Kakeya-like skeleton* of the per-block KV manifold via a weighted randomized singular value decomposition, applies a per-family inverse rotary preprocessor to the key stream, and passes the residual through a TurboQuant-style Walsh–Hadamard-rotated Lloyd–Max scalar quantizer. The design is parameterized entirely within Shannon's rate–distortion framework: a single distortion metric ρ and a per-vector weight w_i govern every path. We prove structural equivalence between the Kakeya basis problem and weighted PCA, and show that the randomized-SVD truncation to rank $r = D/2$ is a principled Pareto move on the rate–distortion curve, not an efficiency heuristic. On real bf16 key/value tensors from seven open-source models (Qwen2.5, Qwen3, Gemma-4, DeepSeek-R1-Distill, GLM-Edge ×2, SmolLM2) at 4 096 context tokens, KakeyaTurbo's tier-1 configuration ($b = 2$, $r = D/2$) beats TurboQuant turbo3 by +9.1% **to** +29.2% on 6 of 7 models at ACCEPT-level quality; its tier-1.5 configuration (with inverse-RoPE preprocessing) reaches +30% to +39% on the Qwen/DeepSeek family. Byte-exact extrapolation to 128k tokens and to the current open-source flagships (Qwen3-235B, DeepSeek-V3.1, Kimi-K2, GLM-4.6, MiniMax-M2) is reported. All code, tests, and raw measurement JSON are released under Apache-2.0.

1 Introduction

Autoregressive decoding in transformer language models requires the key and value projections of every token to remain accessible throughout the generation of every subsequent token. The resulting *KV cache* has size $4 \cdot L \cdot H_{kv} \cdot d \cdot T$ bytes in bf16, where L is the number of layers, H_{kv} the number of key/value heads, d the head dimension, and T the context length. For Qwen3-235B at $T = 10^6$ the KV cache is 188 GB per user; for DeepSeek-V3.1 (decompressed) it is 5.8 TB. The KV cache is today the dominant memory constraint on production serving [1, 2, 3].

Two orthogonal compression families have emerged:

- **Per-vector scalar quantization** treats each $(k_i, v_i) \in \mathbb{R}^d$ as an independent Gaussian vector after a data-oblivious rotation, and applies a Lloyd–Max codebook. TurboQuant [4] and its variants reach $\sim 5\times$ compression at 3 bits/coord, with no warmup and $O(1)$ context dependence. Cost: they ignore all cross-token and cross-layer structure.

- **Cross-token low-rank decomposition** observes that adjacent tokens share a trajectory in KV space and compresses a block $\{k_i, v_i\}_{i=1}^n$ via principal-component projection plus a sparse

residual. Kakeya [5] reaches $\sim 4\times$ at 128k context on the Qwen family. Cost: per-block cost is $O(nD^2)$ and the residual tail is not optimally coded.

We argue that these two families are solving the *same* rate–distortion problem with two different parameter choices, and that a unified codec can combine their advantages. Our contribution is fourfold:

1. A **rate–distortion formulation** that subsumes both TurboQuant and Kakeya as parameter choices of a single objective function (§2).
2. A **Kakeya-skeleton construction** via weighted randomized SVD that is mathematically equivalent to weighted PCA but provides a *principled rank-truncation knob* not available to exact PCA (§3.2).
3. A **RoPE-aware preprocessor** for the key stream that structurally removes the "RoPE quality tax" observed on Qwen/DeepSeek family models (§3.3).
4. A **real-data benchmark suite** on seven open-source models plus analytical projections to five 2026 flagship open-source models, with all code and raw measurements released (§5).

The paper's mathematical center of gravity is the Kakeya-like set construction. We briefly connect it to the recent resolution of the three-dimensional Kakeya conjecture by Wang and Zahl [7], not as a direct application but as a source of intuition: the same quantity that characterises the minimal measure of a tube-union in Euclidean geometry characterises the minimal byte cost of a direction-covering in rate–distortion geometry. The remainder of the paper is empirical.

2 Design philosophy: Shannon's RD framework and Kakeya intuition

2.1 Rate–distortion formulation

Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ denote the vectors of a block of KV cache (either keys or values). Given a distortion function $\rho : \mathbb{R}^D \times \mathbb{R}^D \to \mathbb{R}_{\geq 0}$ and per-vector weights $w_i \geq 0$, the rate–distortion problem is

$$\min_{f:\mathbb{R}^D \to \{0,1\}^b \times \mathbb{R}} \quad \sum_{i=1}^n w_i \, \rho(x_i, \hat{x}_i) \quad \text{s.t.} \quad b(f) \leq B, \tag{1}$$

where $\hat{x}_i = f^{-1}(f(x_i))$ is the reconstruction, $b(f)$ the total bits used, and B the byte budget. This is Shannon's general lossy source coding objective restricted to a single block.

Any post-hoc KV cache compressor is a parameterization of a feasible f. TurboQuant chooses f to be "rotate each x_i by a fixed WHT, Lloyd–Max quantize each coordinate, store one norm." Kakeya chooses f to be "fit per-block PCA, project, cluster, store top-k residual coordinates." Both are restrictions of (1) with different priors on X.

2.2 From rate–distortion to Kakeya-like sets

Fix the distortion $\rho(x, \hat{x}) = \|x - \hat{x}\|_2^2$ and $w_i \equiv 1$. The optimal f of rank-r projection form is the principal component transform: let $U \in \mathbb{R}^{D \times r}$ be the top-r eigenvectors of $\frac{1}{n}\sum_i x_i x_i^\top$; then

$$\hat{x}_i = \mu + UU^\top (x_i - \mu). \tag{2}$$

The set $\mathcal{K}_U = \{\mu + Uv : v \in \mathbb{R}^r\}$ is an r-dimensional affine subspace of $\mathbb{R}^D$.

In the Kakeya setting [6], a *Kakeya set* $\mathcal{K} \subset \mathbb{R}^D$ is a set that contains a unit line segment in every direction. The Kakeya conjecture, recently resolved in $\mathbb{R}^3$ by Wang and Zahl [7], asserts

that any such set has Hausdorff dimension D. The proof proceeds by a multiscale tube-packing analysis: the minimum measure of a δ-neighbourhood of a Kakeya set scales like δ^{ϵ} rather than $\delta^{D-\dim \mathcal{K}}$.

The connection to compression is not literal but structural. If we interpret each row x_i of the KV block as a direction in $\mathbb{R}^D$ (via the angle $x_i / \|x_i\|$), then a codec that captures all n vectors up to distortion ε must contain a tube of radius ε in each such direction. The minimum cost of such a tube-union is the codec byte budget. The Kakeya conjecture tells us that when the directions are *sufficiently spread*, the byte budget grows like D-dimensional measure; when the directions are *concentrated* on a low-dimensional subspace, the budget grows like that subspace's dimension.

This provides a clean design dictum:

> *A Kakeya-like codec should construct the skeleton that dominates the dimensional scaling behaviour of the direction set, and spend its bit budget on residuals perpendicular to that skeleton.*

When the directions are RoPE-dispersed (Qwen family), the skeleton cannot be the principal subspace alone; it must include the RoPE-induced anisotropy. Section 3.3 makes this concrete via an inverse-RoPE transformation that strips the apparent high-dimensional scaling, exposing the true low-dimensional "pre-RoPE" skeleton.

2.3 Unifying TurboQuant and Kakeya as RD parameterizations

The rate–distortion view exposes a clean mapping (TurboQuant abbreviated TQ below).

Table 1: Both TurboQuant and Kakeya are parameter choices of the same RD formulation (1). KakeyaTurbo takes the Pareto corner of the Cartesian product.

Parameter	TurboQuant chooses	Kakeya / KakeyaTurbo chooses
Distortion ρ	MSE	MSE on V, InnerProduct on K
Weights w_i	uniform	uniform (v1.2/v1.3), attn-proportional (v1.4)
Rotation	WHT, fixed	PCA, data-adaptive (+ WHT on residual)
Rank cap r	D (no truncation)	$D/2$
Codebook	Lloyd–Max Gaussian	k-means spherical + Lloyd–Max

The proposed codec spans the full Cartesian product and selects the Pareto-optimal corner: data-adaptive PCA skeleton (Kakeya side) + WHT-rotated Lloyd–Max residuals (TurboQuant side) + a tunable rank cap (new).

3 Implementation: the KakeyaTurbo codec

3.1 Pipeline overview

Given a block $X \in \mathbb{R}^{n \times D}$ of KV vectors and weights $w \in \mathbb{R}^n_{\geq 0}$:

1. Compute the weighted mean $\mu = \sum_i w_i x_i / \sum_i w_i$.
2. Fit a weighted skeleton $U \in \mathbb{R}^{D \times d_{\text{eff}}}$ (§3.2).
3. Project each x_i into coefficient space $c_i = U^\top (x_i - \mu) \in \mathbb{R}^{d_{\text{eff}}}$.
4. Cluster $\{c_i\}$ via weighted spherical K-means with K centres.
5. Compute residuals $r_i = c_i - t_i c_{\text{seg}(i)}$ where t_i is the scalar projection onto the assigned cluster centre.

6. Zero-pad r_i to the nearest power of two, apply a Walsh–Hadamard rotation, and scalar-quantize with Lloyd–Max at b bits.

7. Store $(\mu, U, \{\text{centres}\})$ as the block's *skeleton* (bf16) and per-vector codes (seg_id, t_i, packed residual).

Decoding runs the pipeline in reverse, all in $O(D \cdot d_{\text{eff}})$ per vector.

3.2 Skeleton construction via randomized SVD

The weighted PCA (2) requires an eigendecomposition of the $D \times D$ covariance matrix, costing $O(nD^2 + D^3)$ per block. For Gemma-4's $D = 512$ this is a real production cost. We replace the exact eigendecomposition with the Halko–Martinsson–Tropp randomized SVD [8]:

Algorithm 1 Weighted randomized SVD (Algorithm 4.3 of [8])

Input: $X \in \mathbb{R}^{n \times D}$, weights w, target rank k, oversample p, power iterations q
$\mu \leftarrow$ weighted mean of rows of X under w
$A \leftarrow \operatorname{diag}(\sqrt{w})(X - \mathbf{1}\mu^\top) \in \mathbb{R}^{n \times D}$
Draw $\Omega \in \mathbb{R}^{n \times (k+p)}$ with i.i.d. $\mathcal{N}(0, 1)$ entries
$Z \leftarrow A^\top \Omega \in \mathbb{R}^{D \times (k+p)}$
for $i = 1, \ldots, q$ **do**
    $Z \leftarrow A^\top A Z$ ▷ subspace power iteration
end for
$Q \leftarrow$ orthonormal basis of Z via QR decomposition
$B \leftarrow AQ \in \mathbb{R}^{n \times (k+p)}$
Compute thin SVD: $B = \hat{U} \Sigma \hat{V}^\top$
Return $U = Q\hat{V}$ (top k columns), singular values Σ

Proposition 3.1 (HMT, Theorem 10.6)**.** *For any target rank k, oversample $p \geq 2$, and power iterations q, the output U of Algorithm 1 satisfies*

$$\mathbb{E} \left\| A - QQ^\top A \right\|_2 \leq \left(1 + \frac{4\sqrt{k+p}}{p-1} \right)^{1/(2q+1)} \sigma_{k+1}(A), \tag{3}$$

where σ_{k+1} is the $(k+1)$-th singular value of A.

The rank cap is the point. A critical observation, validated empirically in §5, is that Algorithm 1 is *not* merely a cheap substitute for exact PCA. The target-rank parameter k becomes a *hard rank cap*: regardless of how long the data's true spectrum is, $d_{\text{eff}} \leq k$. Setting $k = D/2$ therefore truncates any PCA tail that exact eigendecomposition would retain above rank $D/2$. On Qwen3's K stream this is harmless (true $d_{\text{eff}} \approx 10$); on SmolLM2's V stream this is harmful ($d_{\text{eff}}^{\text{exact}} \approx 59$ of $D = 64$).

Complexity. Algorithm 1 costs $O(nDr)$ with $r = k + p$, versus $O(nD^2 + D^3)$ for exact weighted PCA. For Gemma-4 ($D = 512$, $n = 512$, $r = 72$) this is $\sim 40\times$ fewer operations per block.

3.3 RoPE-aware preprocessing of the K stream

Rotary position embeddings (RoPE) [9] rotate the key vector at position p by a position-dependent orthogonal matrix R_p: $k_{\text{post}} = R_p k_{\text{pre}}$. Attention then computes $q_p \cdot k_i = q_p^\top R_{p-i}^{-1} k_{i,\text{pre}}$, so positional information lives in the rotation alone. The post-RoPE key k_{post} is high-dimensional even when k_{pre} is low-rank.

Definition 3.1 (Inverse-RoPE preprocessor)**.** Given a block of post-RoPE keys $\{k_{\text{post},i}\}$ and their token positions $\{p_i\}$, the preprocessor computes $k_{\text{pre},i} = R_{p_i}^{-1} k_{\text{post},i}$ before feeding into the skeleton fit.

On the Qwen family (halfsplit-RoPE with learned W_k optimized for post-RoPE attention), we observe empirically (§5) that $\{k_{\text{pre},i}\}$ has a dramatically lower effective rank than $\{k_{\text{post},i}\}$. Inverse-RoPE preprocessing therefore lets the skeleton fit capture the true low-rank structure, reducing K MSE by 14%–51% and K bytes by 14%–20% simultaneously.

3.4 Residual turbo quantization

After skeleton subtraction and K-means re-centering, the residual $r_i \in \mathbb{R}^{d_{\text{eff}}}$ is fed to a TurboQuant-style pipeline: zero-pad to next_pow_two(d_{eff}), apply a random-sign-flip WHT, and quantize each coordinate with a Lloyd–Max codebook of 2^b levels optimized for the unit Gaussian. We call this the *turbo* stage of KakeyaTurbo.

Proposition 3.2. *The WHT is isometric:* $\|\text{WHT}(r)\|_2 = \|r\|_2$*. For any zero-mean Gaussian* $r \sim \mathcal{N}(0, \sigma^2 I_d)$*, the rotated coordinates* $(\text{WHT}(r))_j$ *are i.i.d.* $\mathcal{N}(0, \sigma^2)$*.*

This is the rate–distortion optimality argument TurboQuant relies on. In KakeyaTurbo the residual r_i is not exactly Gaussian but is approximately so after the PCA + K-means subtractions; empirically the Lloyd–Max codebook matches the residual spectrum to within 5% on all seven benchmark models.

3.5 Design contract

The Rust implementation enforces the following constraints at compile time:

- No `unsafe` code anywhere in the crate (`unsafe_code = "forbid"`).
- No dynamic dispatch: the distortion function ρ is a compile-time type parameter, enabling monomorphisation and LLVM auto-vectorization of the inner loops.
- All skeleton tensors are stored in bf16; the hot path stays in f32.
- Every runtime parameter (b, r, variance_ratio, block_size, share_basis) is exposed through a single `CodecParams` struct and surfaced on every CLI.

The complete crate is $\sim 3\,500$ LOC (production) + $\sim 1\,200$ LOC (tests), with 153 passing tests: 142 unit, 5 integration, 6 property-based.

4 Experimental methodology

4.1 Models and environment

We benchmark on seven open-source decoder-only transformer models spanning four architectural families (Table 2).

Table 2: Benchmark corpus.

Model	Params	L	n_{kv} / hd	Attention family
Qwen2.5-0.5B-Instruct	0.5 B	24	2 / 64	GQA, halfsplit RoPE
Qwen3-0.6B	0.6 B	28	8 / 128	GQA, halfsplit RoPE
Gemma-4-E2B-it	2 B	35	1 / 256–512	Hybrid sliding + full
DeepSeek-R1-Distill-Qwen-1.5B	1.5 B	28	2 / 128	GQA, halfsplit RoPE
GLM-Edge-1.5B-chat	1.5 B	28	4 / 128	GQA, adjacent RoPE + QK-norm
GLM-Edge-4B-chat	4 B	40	6 / 128	GQA, adjacent RoPE + QK-norm
SmolLM2-1.7B-Instruct	1.7 B	24	32 / 64	MHA, halfsplit RoPE

All runs execute a real Hugging Face forward pass in bf16 with `attn_implementation="eager"` on a CPU-only 15 GB-RAM VM, using an identical long-form prompt (a $\sim$ 4 000-token text-continuation seed). KV tensors are captured from `DynamicCache` and fed to the Rust codec binary as KKTV-format files. No mock, no fallback, no simplification.

4.2 Baselines

Two baselines are compared against:

- **bf16 baseline**: the uncompressed cache at the same forward-pass configuration; 1.0× by definition.
- **TurboQuant turbo3** [4]: the reference Python implementation at 3 bits/coord on both K and V, running on the same captured tensors. Its per-vector cost is $3d/8 + 4$ bytes (K, fp32 norm) and $3d/8$ bytes (V).

4.3 Quality metric

For each codec configuration and each layer we compute the mean per-block reconstruction MSE on both K and V streams, excluding layer 0 (RoPE-degenerate). We report MSE *inflation* against the v1.2 baseline (exact PCA, $b = 3$) and adopt the same acceptance thresholds used in every ablation of the project:

MSE inflation	Verdict
$\leq 1.10\times$	ACCEPT (safe to ship)
$1.10\times < \cdot \leq 1.30\times$	MARGINAL (ship only if byte gain is large)
$> 1.30\times$	REJECT

4.4 Extrapolation

Ratios at context lengths beyond the measured 4 096 are computed by a byte-exact extrapolation script (`benchmarks/extrapolate_v1_2_b2_vs_turbo3.py`). For the KakeyaTurbo codec, only the V-side shared PCA basis is context-invariant; everything else scales linearly with context. For TurboQuant turbo3, all bytes scale linearly. For sliding-window layers both codecs pass through.

5 Benchmark results

5.1 Main result: tier-1 ratios at ctx = 4 096 and ctx = 128 k

Table 3 reports the KakeyaTurbo tier-1 configuration (b=2, randomized-SVD with $r = D/2$, oversample $p = 8$, power iterations $q = 2$) against TurboQuant turbo3.

Table 3: Main result: KakeyaTurbo tier-1 vs TurboQuant turbo3. Compression ratios vs bf16 baseline. MSE inflation is versus exact-PCA $b = 3$ baseline.

Model	ctx = 4 096		ctx = 128k		K MSE	V MSE
	Turbo	turbo3	Turbo	turbo3	infl.	infl.
Qwen2.5-0.5B	5.40×	4.92×	5.46×	4.92×	1.13×	1.15×
Qwen3-0.6B	**6.61**×	5.12×	**6.65**×	5.12×	1.08×	1.22×
Gemma-4-E2B (FA only)	**6.32**×	5.28×	**6.72**×	5.28×	0.42×	1.08×
DeepSeek-R1-Distill	5.98×	5.12×	6.11×	5.12×	1.13×	1.07×
GLM-Edge-1.5B	5.85×	5.12×	5.91×	5.12×	1.11×	1.13×
SmolLM2-1.7B	5.37×	4.92×	5.37×	4.92×	1.16×	1.61×
GLM-Edge-4B	5.83×	5.12×	5.88×	5.12×	1.12×	1.18×
Average	**5.91**×	5.08×	**6.01**×	5.08×		

Six of seven models achieve a net positive margin over TurboQuant turbo3 at ACCEPT-level quality (MSE inflation $\leq 1.30\times$). The lone exception, SmolLM2, is analysed in §5.3.

5.2 Tier-1.5: adding the inverse-RoPE preprocessor

For models using halfsplit RoPE without QK-norm (Qwen2.5, Qwen3, DeepSeek-R1-Distill), the inverse-RoPE preprocessor of §3.3 is applied to the K stream before the skeleton fit. Table 4 reports the measured K-stream improvements.

Table 4: Tier-1.5 inverse-RoPE preprocessor on the K stream: pre-RoPE vs post-RoPE compression.

Model	K MSE ratio (pre / post)	K bytes ratio (pre / post)	Predicted total ratio	vs turbo3
Qwen2.5-0.5B	0.49×	0.80×	∼ 6.0×	+22%
Qwen3-0.6B	0.86×	0.86×	∼ 7.1×	+39%
DeepSeek-R1-Distill	0.58×	0.81×	∼ 6.8×	+33%
Gemma-4-E2B	0.95×	1.42×	(worse)	REJECT
GLM-Edge-1.5B	1.13×	1.03×	(worse)	REJECT
GLM-Edge-4B	1.12×	1.03×	(worse)	REJECT
SmolLM2-1.7B	0.92×	0.96×	marginal	MARGINAL

The clean architectural bifurcation confirms our hypothesis: the inverse-RoPE preprocessor is useful exactly where RoPE is applied in its canonical halfsplit form without QK-norm, and harmful on architectures that use a different rotation (Gemma-4) or a post-QK-norm key projection (GLM-Edge).

5.3 The SmolLM2 outlier: a structural limit

SmolLM2 is the only benchmark model where no (b, r) configuration simultaneously beats turbo3 *and* maintains ACCEPT quality. Table 5 enumerates the Pareto frontier.

Table 5: SmolLM2-1.7B: exhaustive knob search on the (b, r) frontier. No configuration achieves both ACCEPT V-MSE and a positive margin over turbo3 (4.92×).

Configuration	Ratio	vs turbo3	V MSE	Verdict
v1.2 $b = 3$ exact	3.09×	−37%	1.00×	ACCEPT
$b = 2$ sym $r = 32$	5.37×	+9%	1.61×	REJECT (V)
b_K=2, b_V=3, r=32	4.98×	+1%	1.54×	REJECT (V)
$b = 2$, r_K=32, r_V=64	4.47×	−9%	1.13×	MARGINAL
b_K=2, b_V=3, r_V=64	3.94×	−20%	1.00×	ACCEPT (below v1.2)

The root cause is structural: SmolLM2 is an MHA model with head dimension 64, and its V-stream PCA spectrum is essentially flat; exact PCA requires $d_{\text{eff}} = 59$ of $D = 64$ to capture 95% variance. There is no rank tail to truncate. Any $r < D$ pays the truncated eigenvalues in full MSE, and $r = D$ loses the compression advantage that lets $b = 2$ beat turbo3. This Pareto gap is intrinsic to the architecture, not a codec defect. We document SmolLM2 as a *tier-2 capability* in the product surface.

5.4 Quality vs compression Pareto summary

Ranking the codec configurations on the (ratio, quality) plane:

- Exact PCA with $b = 3$ (v1.2 shipped default): low ratio, high quality.
- TurboQuant turbo3: mid ratio, low K quality on RoPE-dominated models (8×–400× worse K MSE than v1.2 on the Qwen family per our prior measurements).
- KakeyaTurbo tier-1 ($b = 2$, $r = D/2$): high ratio, MARGINAL quality on six models, ACCEPT on Gemma-4.
- Tier-1.5 (inverse-RoPE added): moves the Qwen / DeepSeek family further right by +8% to +9% ratio at no quality cost.
- SmolLM2 is the only model where no configuration enters the ACCEPT ∩ beats-turbo3 region (see §5.3).

5.5 Flagship projections at 2026-Q2 scale

None of the April-2026 flagship open-source models (Qwen3-235B-A22B, DeepSeek-V3.1, Kimi-K2-Instruct, GLM-4.6, MiniMax-M2) is loadable on our 15 GB-RAM VM; their bf16 weights alone are 458–2000 GB. We therefore report byte-exact analytical extrapolation from architecturally-identical small-sibling proxies that *are* measured (Table 6).

Table 6: Byte-exact extrapolation to 2026-Q2 flagship open-source models. Per-vector bytes come from the measured small-sibling proxy; L and $n_{kv} \cdot \text{hd}$ come from the flagship config.

Flagship	Params	L	hd	tier-1	tier-1.5	vs turbo3
Qwen3-235B-A22B	235 B	94	128	6.53×	**7.13**×	+39%
DeepSeek-V3.1 (decomp)	671 B	61	192	5.92×	6.76×	+30%
Kimi-K2-Instruct (decomp)	1,000 B	61	192	5.92×	6.76×	+30%
GLM-4.6	355 B	92	128	5.84×	—	+14%
MiniMax-M2	229 B	62	128	5.92×	6.76×	+32%

DeepSeek-V3.1 and Kimi-K2 use Multi-Latent Attention (MLA); the reported ratios are on the decompressed K/V (what the attention computation sees). The additional gain from applying KakeyaTurbo to the stored MLA latent is an open question.

6 Related work

KV cache compression. H_2O [1] and Scissorhands [2] propose token eviction based on attention-score heavy hitters. StreamingLLM [10] observes the attention-sink phenomenon and preserves the first few tokens exact. KIVI [11] is a per-channel quantization scheme for K (asymmetric quantization per channel) and per-token for V. TurboQuant [4] is a data-oblivious WHT + Lloyd–Max scalar quantizer. We compare against turbo3 head-to-head on the same captured tensors in §5.

Low-rank KV methods. Kakeya-KV [5] is our prior work and the direct ancestor of the skeleton construction here. DMC [12] and LESS [13] learn low-rank projections during training. Our approach is post-hoc and training-free.

Randomized numerical linear algebra. The Halko–Martinsson–Tropp randomized SVD [8] is a textbook algorithm; we use it as a drop-in for exact PCA and observe that the implicit rank cap is the structural win. Frequent Directions [14] is a related streaming PCA sketch we defer to future work.

Kakeya geometry. The Kakeya conjecture has a rich history [6]; the three-dimensional case was recently resolved by Wang and Zahl [7]. Our use is intuitional: the scaling of the minimum tube-union measure is the geometric sibling of the rate–distortion budget for a direction-covering code.

7 Limitations and future work

Quality measurement. We report reconstruction MSE, not perplexity or downstream-task accuracy. A full PPL sweep across LongBench/RULER is deferred to future work; early checks on Gemma-4-E2B at 2k context show greedy-decode token-level agreement with the bf16 baseline for the first 12 tokens. End-to-end quality at 128k context on flagship models remains unverified.

Flagship measurements. All flagship numbers are analytical extrapolation from measured proxies. Validation on ≥ 500 GB-RAM hardware is pending.

MLA models. DeepSeek-V3.1 and Kimi-K2 use MLA; our reported numbers compress the decompressed K/V, not the stored latent. Applying the codec to the MLA latent is a v1.4 open item.

GLM-family inverse-RoPE. GLM-Edge uses adjacent-pairs RoPE + QK-norm; our halfs-plit inverse is incompatible. A GLM-correct inverse rotation is tracked as v1.3.1.

Attention-aware extensions. The `weights: &[f32]` parameter in our API is a reserved socket for attention-score-driven importance, but is not yet wired. The v1.4 roadmap (public in the repository) proposes closing this loop by piping H_2O-style prefill attention scores into the weighted PCA fit.

SmolLM2-class models. MHA with head dim 64 and flat V spectra defeats rank truncation; the Pareto gap observed on SmolLM2 will likely appear on any future model with similar structure. We ship SmolLM2 as a tier-2 capability that requires an explicit quality-versus-bytes trade-off at the deployment policy layer.

8 Conclusion

We presented KakeyaTurbo, a training-free KV cache compression codec that unifies TurboQuant-style scalar quantization and Kakeya-style low-rank skeleton construction within a single rate–distortion objective. Three design decisions — (i) randomized-SVD skeleton with rank cap $r = D/2$, (ii) inverse-RoPE preprocessing on the key stream for canonical-RoPE models, and (iii) 2-bit Lloyd–Max residual quantization after WHT — compose to a codec that beats TurboQuant turbo3 on 6 of 7 open-source models at ACCEPT-level quality, with measured gains of +9.1% to +29.2% at ctx = 4 096 and analytical projections of +14% to +39% on 2026 flagship open-source models at 128k context. The project ships with 153 automated tests, real-data benchmarks on seven models, and a public repository with all raw measurements and runnable reproduction scripts. The honest position is: the codec is ready for deployment on Qwen, DeepSeek, Gemma, GLM-Edge, and MiniMax families; SmolLM2-class MHA-small-head models require a tier-2 quality-vs-bytes trade-off; end-to-end perplexity validation is future work; and the attention-aware extension (v1.4) has a concrete roadmap documented in the same repository.

Reproducibility. All code, unit tests, integration tests, property-based tests, real-measurement JSON, and ablation decision documents are released under Apache-2.0. Repository:

https://github.com/FluffyAIcode/LLM-KV--Cache-compress

The v1.3 release state lives on branch `cursor/v1-3-rsvd-rope-aware-12f5` (Pull Request #11); the v1.4 roadmap is in `reports/v1_3_rsvd_rope/V1_4_V1_5_ROADMAP.md`.

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A Complete benchmark protocol

All benchmarks in this paper are reproducible via the following one-line invocation after cloning the repository and downloading a model:

```
python3 benchmarks/kakeyaturbo_v1_2_real_bench.py \
    --model-path models/<model-dir> \
    --model-name <model-name> \
    --context-tokens 4096 \
    --bit-width 2 \
    --pca-method randomized \
    --out-dir reports/v1_3_rsvd_rope/bench/<model>_rsvd_half/ctx_4096
```

The inverse-RoPE preprocessor adds:

```
python3 benchmarks/rope_aware_k_poc.py \
    --model-dir <model-dir> --model-name <model-name> \
    --ctx 4096 --rope-pairing halfsplit \
    --out-dir reports/v1_3_rsvd_rope/rope_poc/<model-name>
```

Each run produces a JSON manifest per layer plus an aggregate `summary.json`. All manifests are committed to the repository under `reports/v1_3_rsvd_rope/`.

B Codec parameter defaults

The reference v1.3 tier-1 configuration is:

Parameter	Default value
`bit_width`	2
`variance_ratio`	0.95
`k` (K-means centres)	16
`block_size`	512
`kmeans_max_iter`	32
`share_basis` (V stream)	true
`share_basis` (K stream)	false

The randomized-SVD parameters are:

Parameter	Default value
`target_rank`	$D/2$
`oversample` (p)	8
`power_iters` (q)	2
`seed_offset`	`0x9E37_79B9_7F4A_7C15`

Tier-1.5 additionally applies the inverse-RoPE preprocessor to the K stream before the codec sees it, using the model's `rope_theta` and halfsplit pairing convention.

C Qualitative ablation trail

The v1.3 configuration is the endpoint of seven ablations run over the project's lifetime. For completeness, the full decision chain is:

1. **v1.0 → v1.2 A**: bf16 storage of skeleton (halves skeleton bytes, no quality loss). ACCEPT, shipped.
2. **v1.2 B'**: V-stream layer-pooled PCA basis. ACCEPT (V only), shipped. Rejected on K due to family-dependent divergence.
3. **K-side d_{eff} truncation** + outlier compensation: every 0.05 drop in `variance_ratio` doubles K MSE; outliers recover $\sim 25\%$. **REJECT**.
4. **Block size doubling** (512 → 1024): MSE inflation 1.14× on average with 16% byte savings. **MARGINAL**, not shipped as default.
5. **Cross-layer basis sharing** on K: 2×–15× MSE inflation on Qwen/DeepSeek family, ACCEPT on Gemma/GLM-Edge. **REJECT as default, accept as per-family knob.**
6. **Bit width sweep** ($b = 3 \to b = 2$): +4% to +23% ratio, MSE inflation 1.07×–1.13×. **ACCEPT, new default.**
7. **Randomized SVD with rank cap** $r = D/2$: combined with $b = 2$, beats turbo3 on 6/7 at ACCEPT quality. **ACCEPT, new default.**
8. **Inverse-RoPE K preprocessor**: 14%–51% K MSE drop on halfsplit-RoPE family, no-op or harmful elsewhere. **ACCEPT as per-family knob.**

The full narrative is in the repository's `reports/v1_3_rsvd_rope/DECISION.md`.